



F-14 TOMCAT



AV-8B HARRIER II



F-35 LIGHTNING II



B-2 SPIRIT



A-10 THUNDERBOLT II



DESIGNED TO BE
INVISIBLE TO RADAR,
THE NIGHTHAWK WAS
THE WORLD'S FIRST
"STEALTH" AIRCRAFT

PASSENGER PLANES

The first commercial flight took place in 1914, with just one paying passenger. Today's largest airliner, the Airbus A380, can carry up to 853 passengers.



DOUGLAS DC3



LOCKHEED L-1049 G SUPER
CONSTELLATION



DE HAVILLAND DH106 COMET 4C



FOKKER F27 FRIENDSHIP



CONCORDE



BOEING 737-800



AIRBUS A380

PRIVATE AIRCRAFT

A range of light, propeller-driven aircraft are privately flown for both pleasure and transport. Some business travellers use private, often luxurious jets as air taxis.



BOEING STEARMAN
MODEL 75



AUSTER J/1
AUTOCRAT



GIPSY MOTH



PITTS S1-S



CESSNA F177RG
CARDINAL



BOMBARDIER LEARJET 45

HELICOPTERS

The rotating blades of helicopters allow them to take off and land vertically, hover, and fly in all directions. These agile aircraft can land in small spaces and reach isolated locations.



BELL JETRANGER



BELL UH-1D
IROQUOIS



WESTLAND
WS-61 SEA KING



WESTLAND WESSEX



SIKORSKY UH-60 BLACK HAWK



WESTLAND
LYNX

BREAKING THE SOUND BARRIER

Supersonic jets flying faster than the speed of sound create a shock wave called a "sonic boom". To slice through the air at such incredible speed, these planes need a slim body, thin wings, and a sharp nose.



1 SLOWER THAN SOUND
Ordinary planes trail behind their own sounds so you can hear them coming.

2 AT THE SPEED OF SOUND
As a supersonic plane nears the speed of sound, sound waves bunch together to form a shock wave.

3 FASTER THAN SOUND
Shock waves trail behind the plane, creating a loud sonic boom.

HOW HELICOPTERS FLY

The rotor blades on a helicopter are like spinning plane wings. As each blade rotates, air is forced over its curved surface and pushed down. This produces the upward force called "lift".



Tilted blade produces more lift for takeoff

TAKEOFF

When the lift produced by the tilted blades is greater than the aircraft's weight, the helicopter rises.



Flatter blade produces less lift while landing

LANDING

The helicopter descends when the lift produced by the flattened blades is less than the aircraft's weight.



Tilted blade in rear produces more lift

Flatter blade in front produces less lift

STRAIGHT AHEAD

When the rotor is tilted forwards, the resulting thrust propels the helicopter forwards.